

WiMAX Testing, Step by Step

Based on orthogonal frequency-division multiple-access (OFDMA) and multiple-input multiple-output (MIMO) technologies, WiMAX systems require specialised test equipment from design to deployment stage

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Worldwide interoperability for Microwave Access (WiMAX) is synonymous to wireless broadband for high-speed Internet. It provides high-speed data transmission on wireless networks in modes like point-to-multipoint (fixed WiMAX) and fully-mobile Internet access (mobile WiMAX). To achieve high-speed data transmission, it is essential that the system meets the specified standards for quality service.

WiMAX Forum has specified certain standards and certifications for WiMAX devices like customer premise equipment, mobile stations and base stations. Although all the devices comply with these standards at the time of designing, errors may creep in during manufacturing. Therefore to enhance the quality of service and efficiency, WiMAX devices and protocols must undergo rigorous tests for certifications before these are deployed in the field.

WiMAX is the first standard that adopted the next-generation OFDMA technology. This modulation scheme is entirely different from the existing ones in the cellular segment. It transmits much more data on multiple carriers than single-carrier transmissions.

Need for specialised equipment

Designing a system with multiple carriers and testing it is the biggest challenge for the design engineers. The increased complexity in the signals makes high-speed measurements very critical. You need some unconventional equipment to generate and analyse these complex signals.

OFDMA combines frequency-division



WiMAX antenna

duplex and time-division duplex to provide varying bandwidth and transmission time to every user. This helps in efficient resource management, but at the same time adds to the complexity of the physical layer. The receiver has to decompose the mixed signal into independent signals. This imposes a big challenge for test and measurement (T&M) at the physical layer level.

The move from single-carrier to multiple-carrier (from single-input single-output to multiple-input multiple-output) technology complicates the system further as this requires multiple antennae for transmission and reception. To test a MIMO system, multiple signal generators and an integrated signal analyser are needed to measure, display and analyse.

One of the greatest challenges before test engineers is synchronisation. Multiple signals require proper synchronisation of time and phase to

get the correct information. Moreover, signal generators should offer high precision in order to make accurate and repeatable measurements.

One of the important considerations is the bandwidth of the test equipment. Various channel bandwidths defined in mobile WiMAX standards are 1.25, 5, 10 and 20 MHz. So the equipment should test at least 20 MHz of the wide-band signal. The numbers of fast Fourier transform (FFT) sub-carriers allowed for signalling are 128, 512, 1024 and 2048. That means the T&M equipment should also be capable of supporting many sub-carriers in the system.

WiMAX test equipment from lab to deployment

WiMAX is perhaps the most enigmatic wireless system deployed in the field. It takes the advantage of higher-order modulation scheme provided by OFDMA and optimised performance provided by MIMO technology. Consequently, a complete set of T&M solutions capable of handling radio frequency, multiple channels and high-speed digital signal processing is required.

WiMAX systems are built around a WiMAX base station and WiMAX mobile stations (receivers). WiMAX Forum has released RF and protocol conformance standards for base stations and mobile stations. Any device carrying WiMAX Forum emblem has to undergo conformance tests using conformance test equipment. "Once the design is verified, every individual module has to undergo tests during the manufacturing stage to meet the standards. Manufacturing tests are a small subset of conformance tests," says Sonali Sarpotdar, application engineer, Agilent Technologies.